Sheet: VeraModule

VERA FPGA

File: vera-fpga.sch

Sheet: BusDecoder

BUS DECODER

File: busdecoder.sch

Sheet: Vera FPGA flash

VERA SPI FLASH
SD CARD INTERFACE

File: vera-fpga-flash.sch

Sheet: CartridgeInterface

CARTRIDGE INTERFACE

File: cartridgeInterface.sch

Sheet: PowerSupply

POWER SUPPLY

File: powersupply.sch

Sheet: VGA Analog

VGA ANALOG

File: vga-analog.sch

FIDUCIAL TOP

FID3 Fiducial

FID4 Fiducial

FIDUCIAL BOTTOM

FID2 Fiducial

FID1 Fiducial

LOGO1
VERA X16 LOGO

LOGO2
C64 READY

LOGO3
KiCAD DESIGN

LOGO5
ESP32_BOBOARD

LBL1

H1 MountingHole

H2 MountingHole

H3 MountingHole

H4 MountingHole

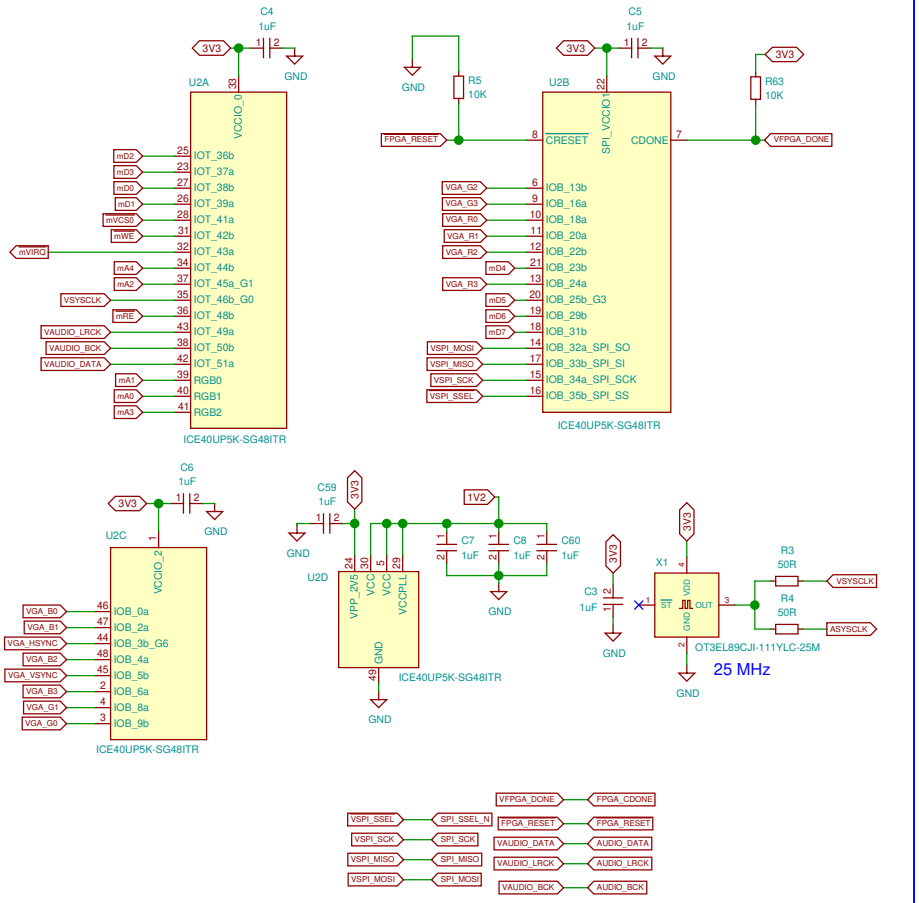
Gianluca Renzi
RetroBit Lab

Sheet: /
File: VERA-MODULE-RBL.sch

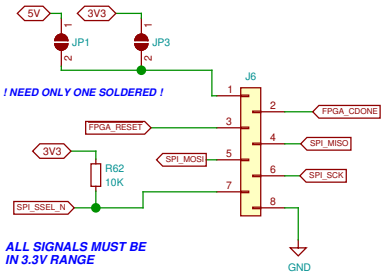
Title: VERA FPGA Audio & Video Board

Size: A4	Date: 2026-10-26	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261104-14:44Z		Id: 1/7

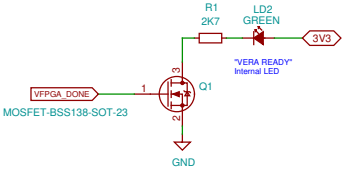
FPGA VERA LOGIC VIDEO & AUDIO CARD



iceprog programmer USB FTDI / SPI

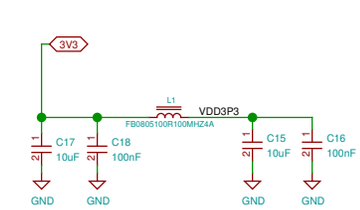


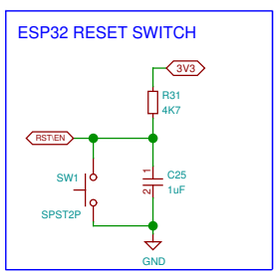
VERA FPGA PROGRAMMED OK

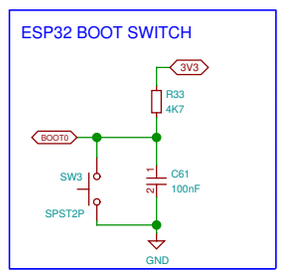


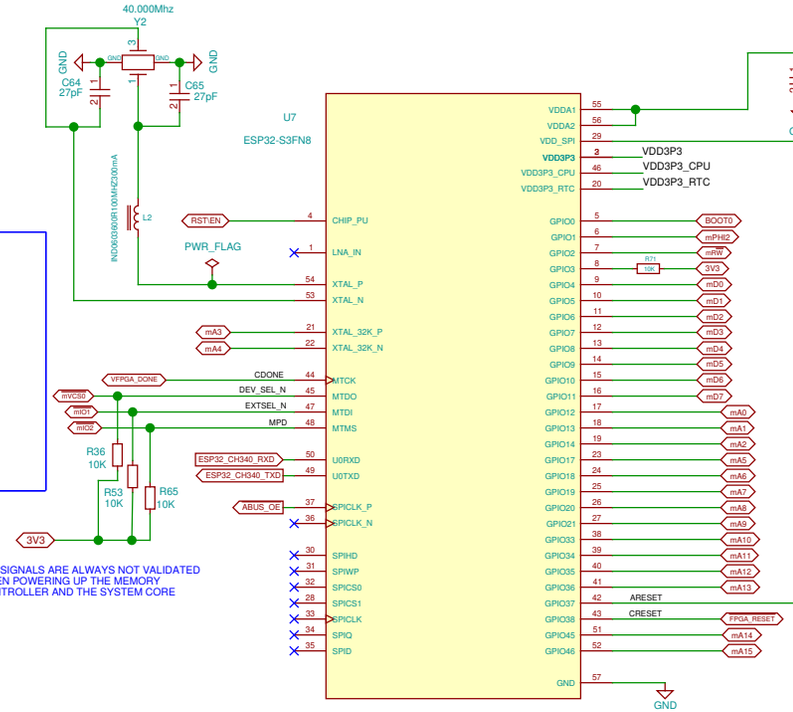
Gianluca Renzi	
RetroBit Lab	
Sheet: /VeraModule/	
File: vera-fpga.sch	
Title: VeraModule	
Size: A4	Date: 2026-10-26
Plotted by eeshow 3a954ac+ 20261104-14:44Z	Rev: 1.0
	Id: 2/7

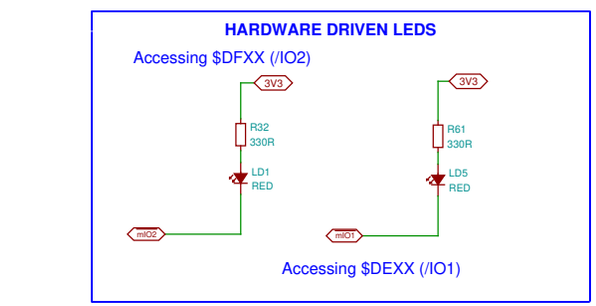
VERA SELECT: latch $DFFD, software only



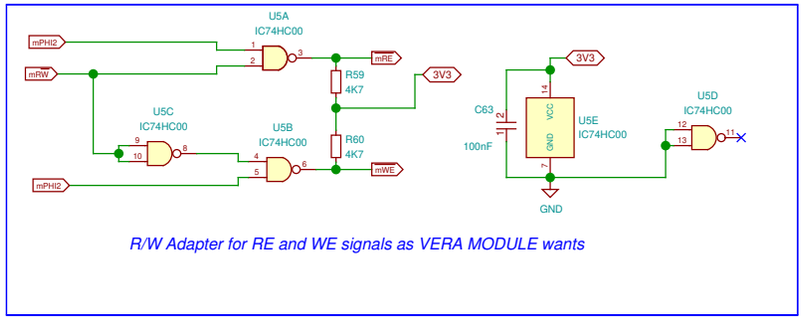


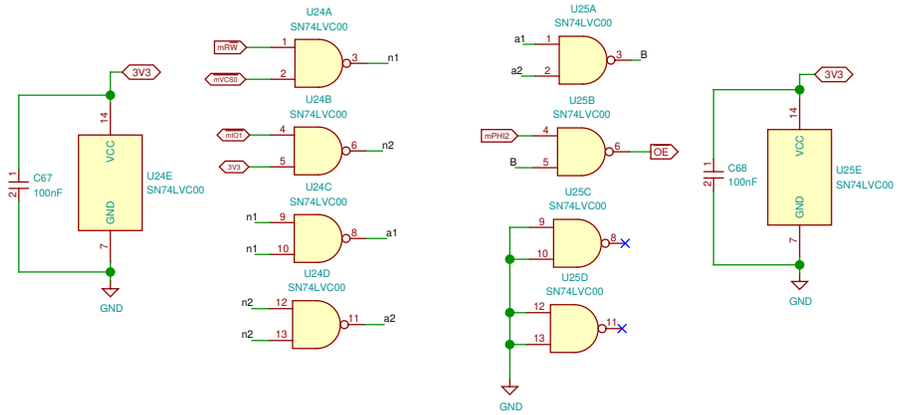






C64 Expansion Port Decoder:
/IO1 GeoRAM window ($DE00-$DEFF),
/IO2 VERA registers ($DF00-$DF1F), latch and GeoRAM registers ($DFFD-$DFFF)





/IO1 ($DE00-$DEFF) -> GeoRAM window, data driven by the ESP32 (U19 enabled by /IO1)
/IO2 $DF00-$DF1F and VERA selected through $DFFD -> mVCS0 (DEV_SEL_N) active
/IO2 $DFFD/$DFFE/$DFFF -> VERA select latch, GeoRAM page/block (write only)

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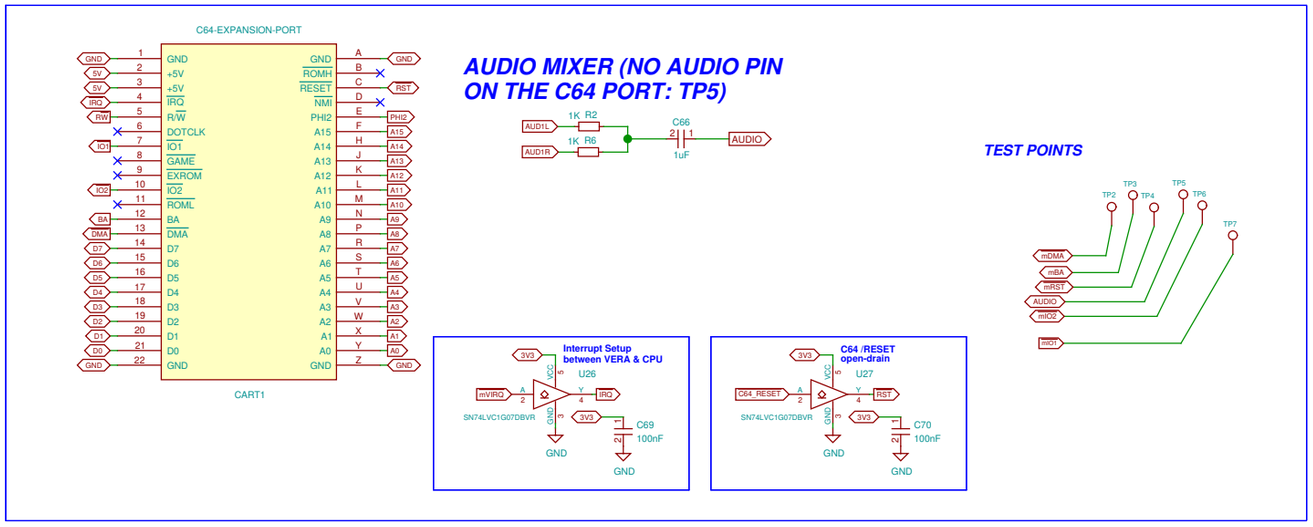
Sheet: /BusDecoder/
File: busdecoder.sch

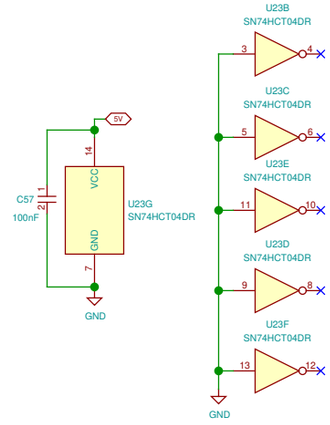
Title: BusDecoder

Size: A3	Date: 2026-10-26	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261104-14:44Z		Id: 3/7

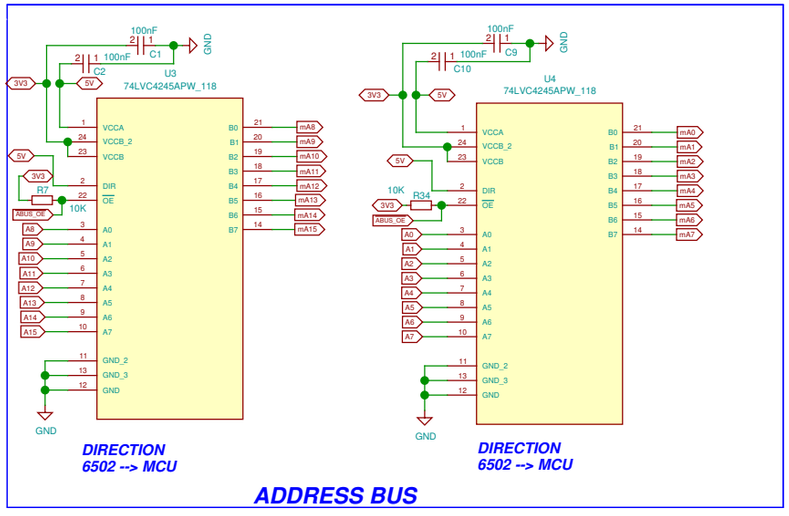
COMMODORE 64 EXPANSION PORT INTERFACE

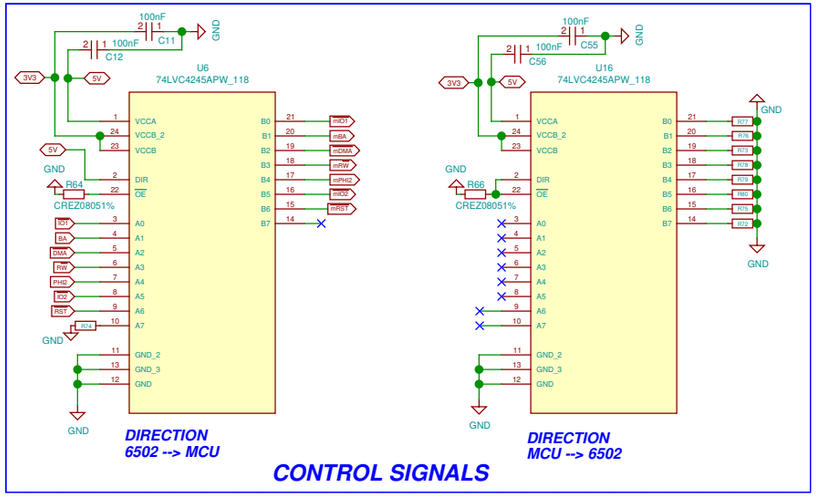
All 8-Bit signals must be shifted from 5V to 3.3V and vice versa

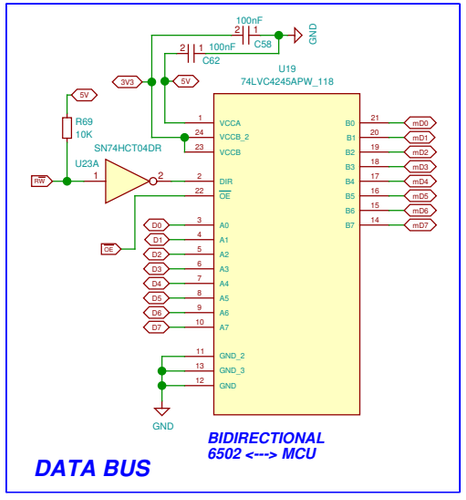




BUS LOGIC LEVEL SHIFTERS

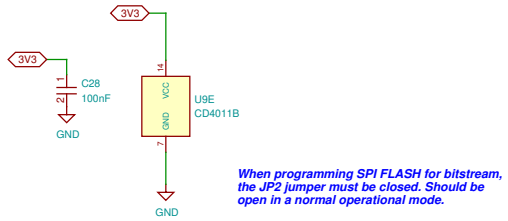




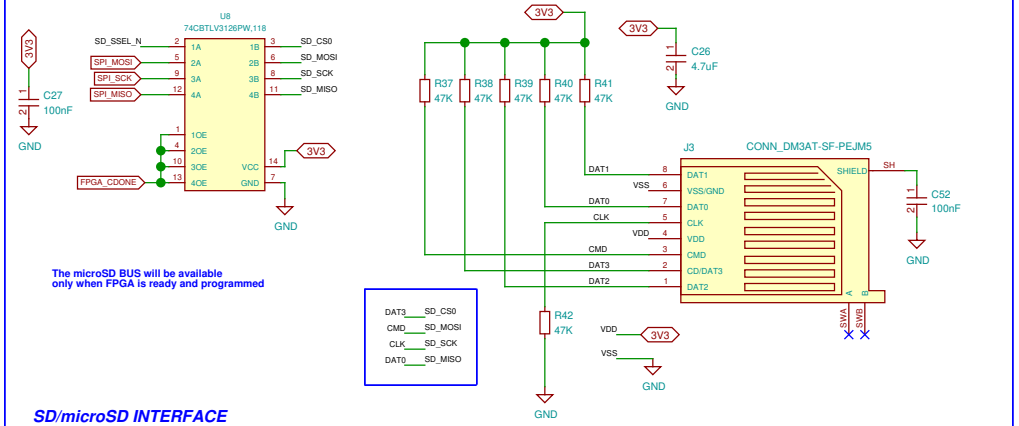


Gianluca Renzi		
RetroBit Lab		
Sheet: /CartridgeInterface/		
File: cartridgeInterface.sch		
Title: CartridgeInterface		
Size: A3	Date: 2026-10-26	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261104-14:44Z		Id: 4/7

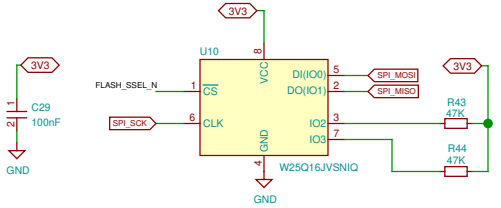
FPGA/SSD Flash Glue Logic



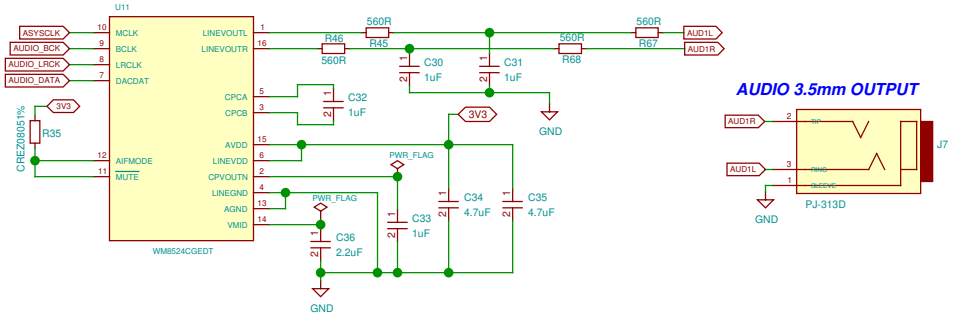
SD/microSD INTERFACE



SPI 16MB FLASH



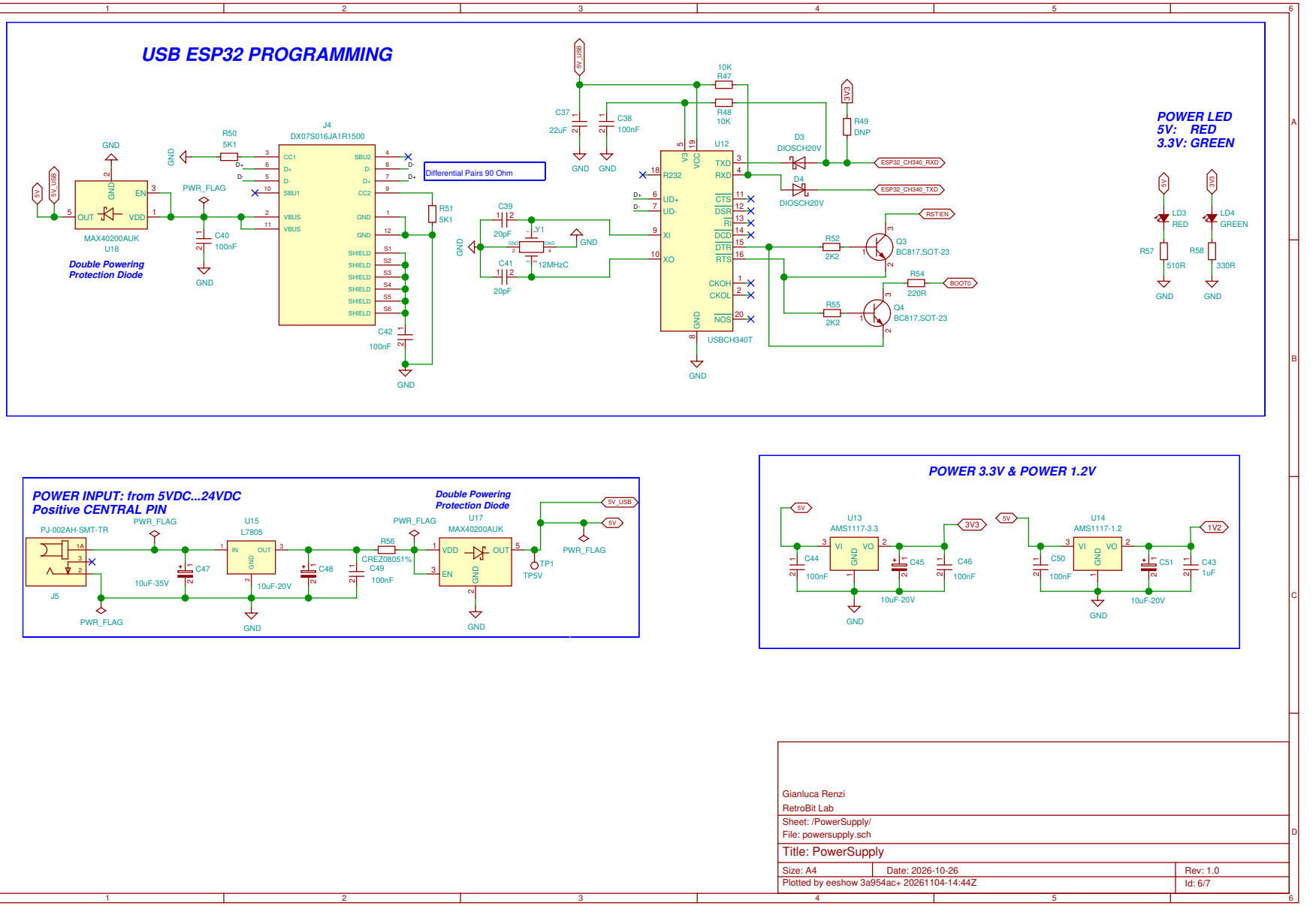
IC DAC/AUDIO 24BIT 192K 16TSSOP



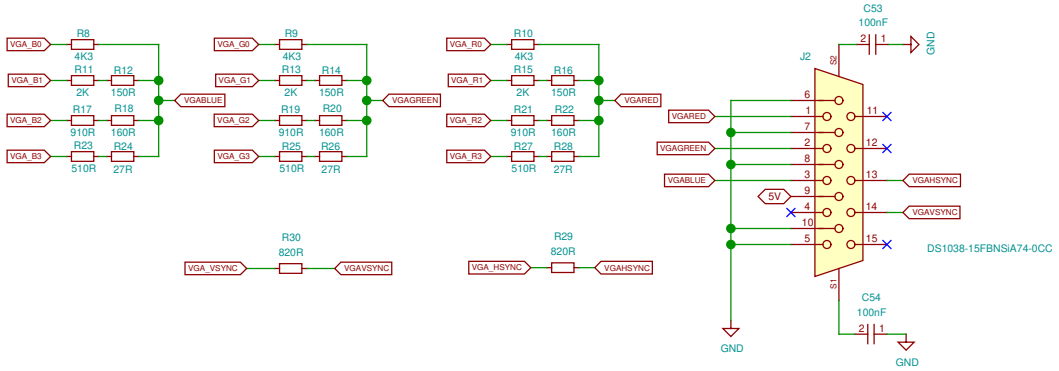
AUDIO 3.5mm OUTPUT

AUDIO SECTION

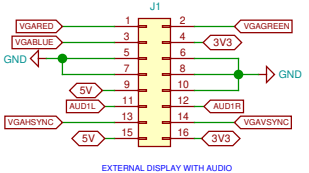
Gianluca Renzi	
RetroBit Lab	
Sheet: /Vera FPGA flash/	
File: vera-fpga-flash.sch	
Title: Vera FPGA flash	
Size: A4	Date: 2026-10-26
Rev: 1.0	
Plotted by eeshow 3a954ac+ 20261104-14:44Z	Id: 5/7


USB ESP32 PROGRAMMING
POWER LED
5V: RED
3.3V: GREEN
Differential Pairs 90 Ohm
Double Powering
Protection Diode
POWER INPUT: from 5VDC...24VDC
Positive CENTRAL PIN
Double Powering
Protection Diode
POWER 3.3V & POWER 1.2V
MAX40200AUK
U18
DX07S016JA1R1500
J4
USBCH340T
U12
AMS1117-3.3
U13
AMS1117-1.2
U14
L7805
U15
MAX40200AUK
U17
PJ-002AH-SMT-TR
J5
BC817,SOT-23
Q3
Q4
DIOSCH20V
D3
D4
12MHzC
Y1
CREZ08051%
TP5V
TP1
ESP32_CH340_RXD
ESP32_CH340_TXD
RST'EN
BOOT0
Gianluca Renzi
RetroBit Lab
Sheet: /PowerSupply/
File: powersupply.sch
Title: PowerSupply
Size: A4
Date: 2026-10-26
Rev: 1.0
Plotted by eeshow 3a954ac+ 20261104-14:44Z
Id: 6/7

ANALOG VGA SIGNALS



EXTERNAL VIDEO CONNECTOR



Gianluca Renzi		
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Sheet: /VGA Analog/		
File: vga-analog.sch		
Title: VGA Analog		
Size: A4	Date: 2026-10-26	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261104-14:44Z		Id: 7/7